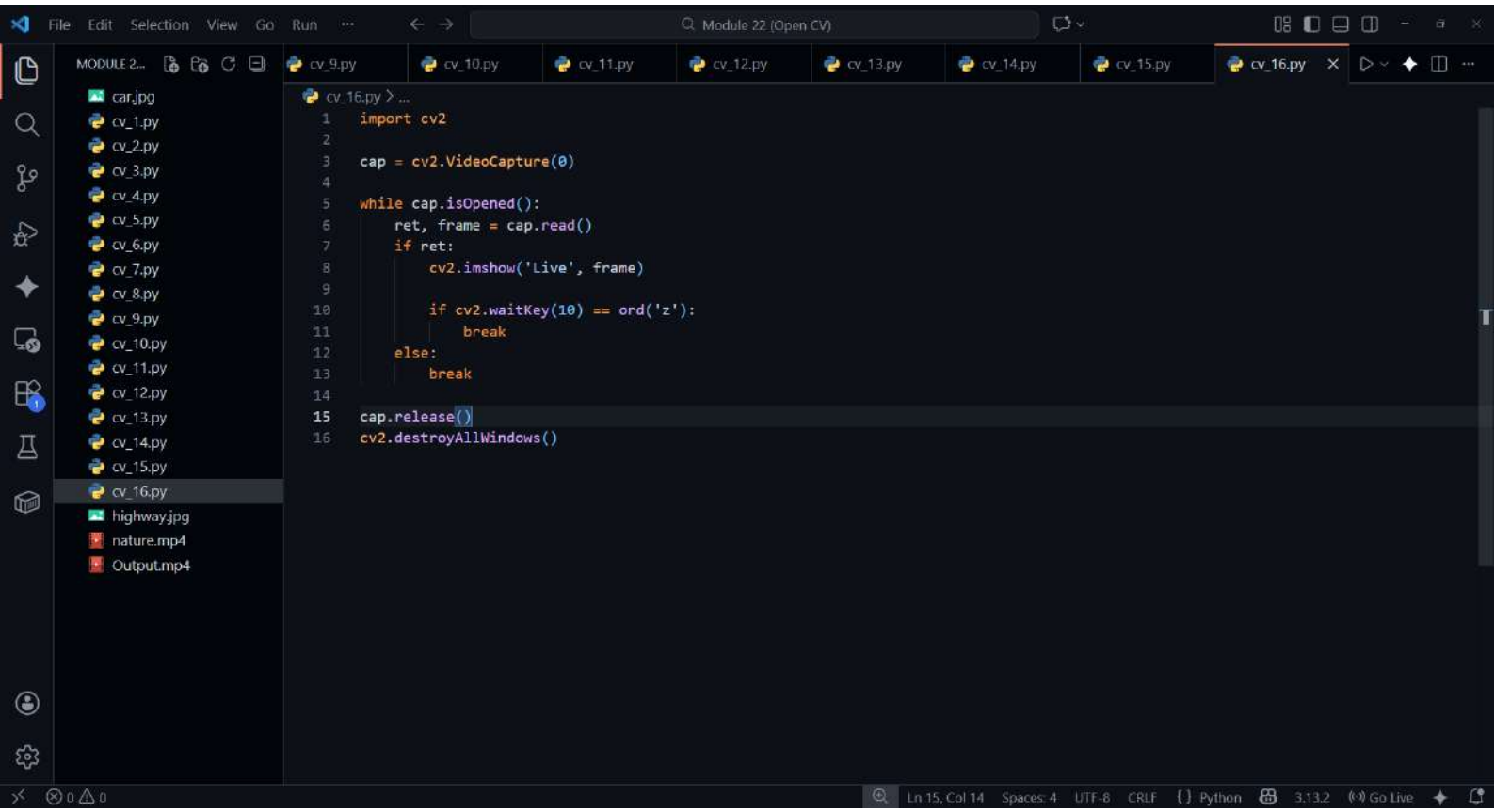
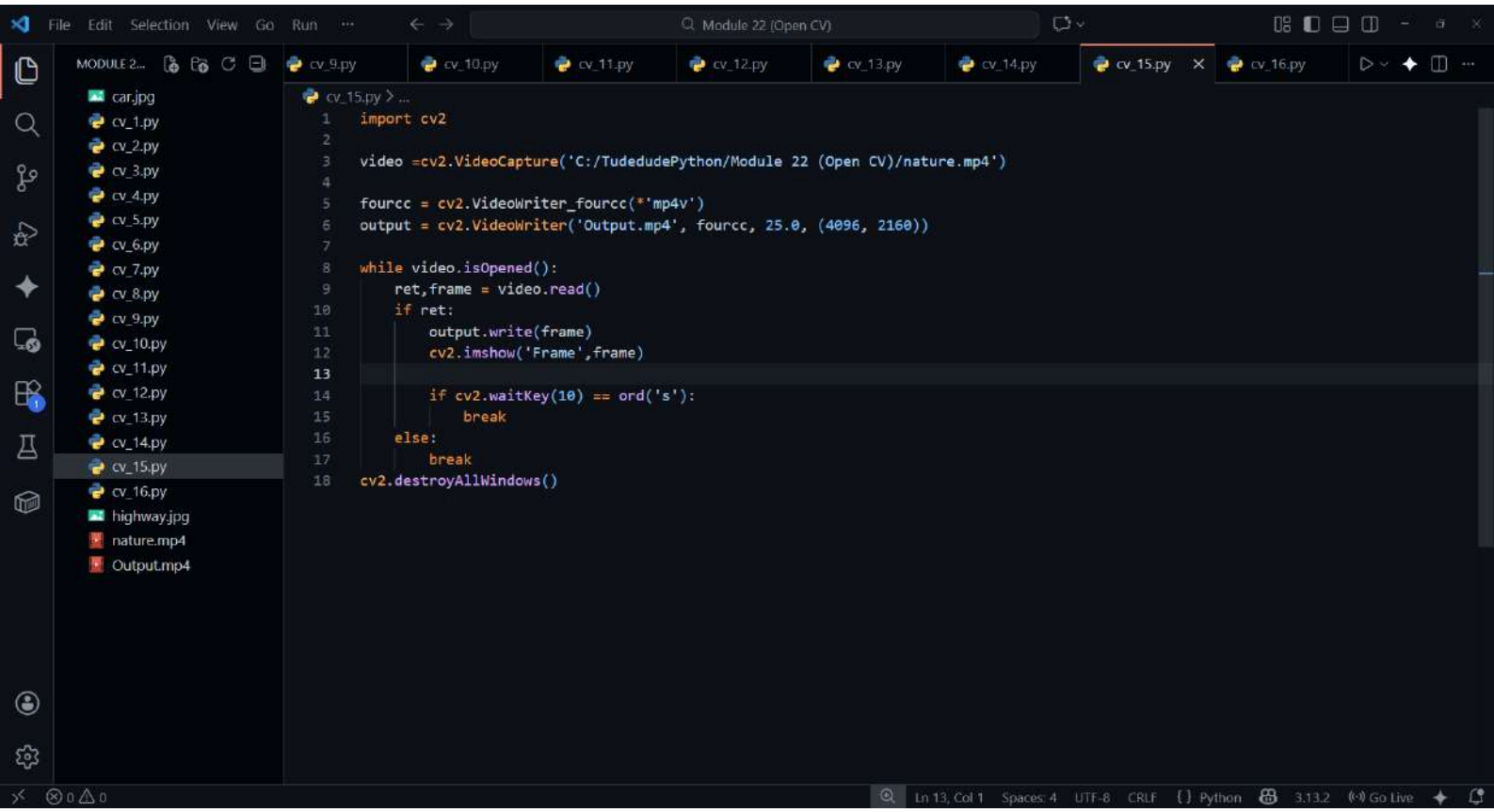

File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_16.py > ...
1 import cv2
2
3 cap = cv2.VideoCapture(0)
4
5 while cap.isOpened():
6 ret, frame = cap.read()
7 if ret:
8 cv2.imshow('Live', frame)
9
10 if cv2.waitKey(10) == ord('z'):
11 break
12 else:
13 break
14
15 cap.release()
16 cv2.destroyAllWindows()
Ln 15, Col 14 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
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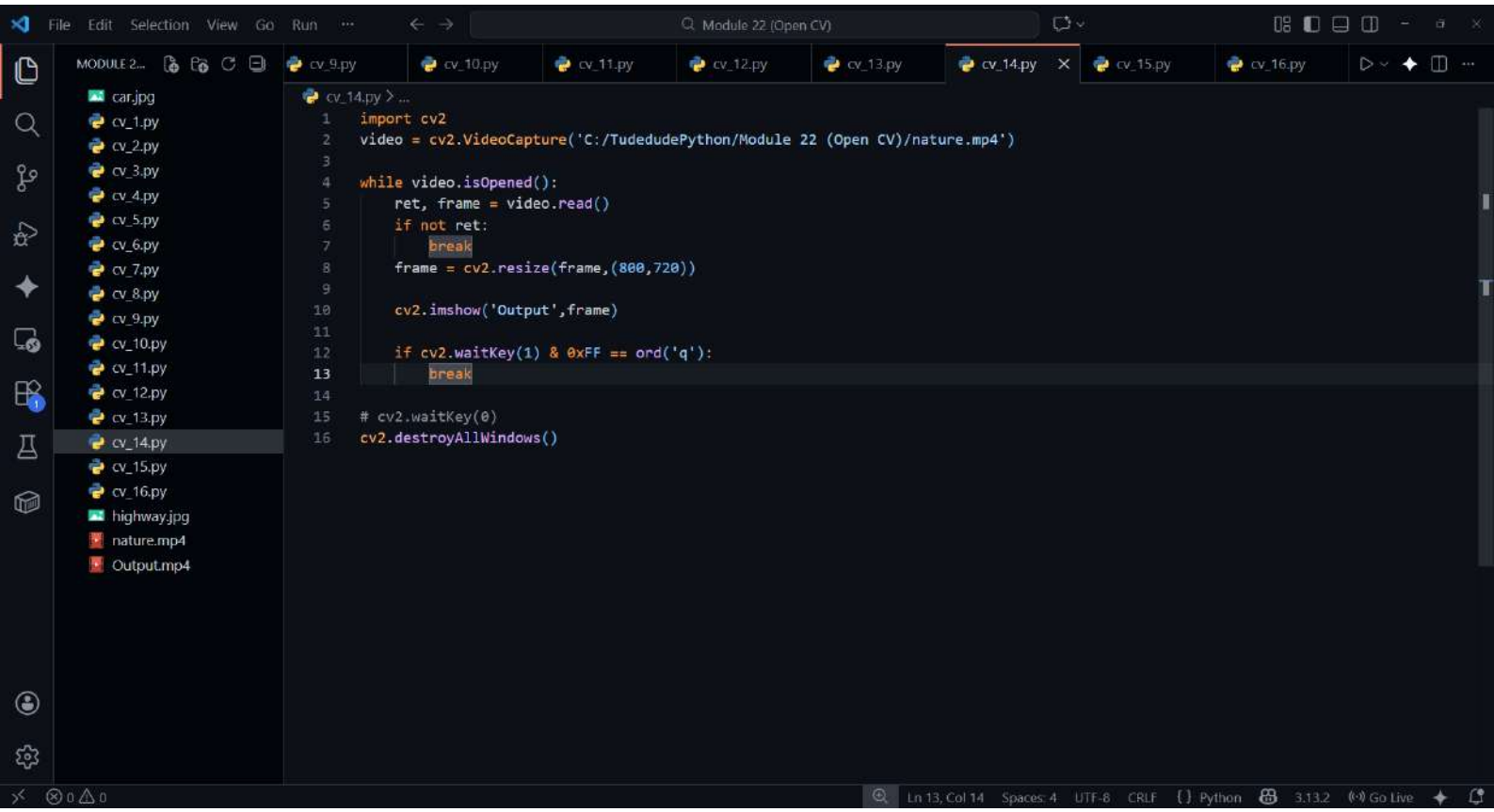
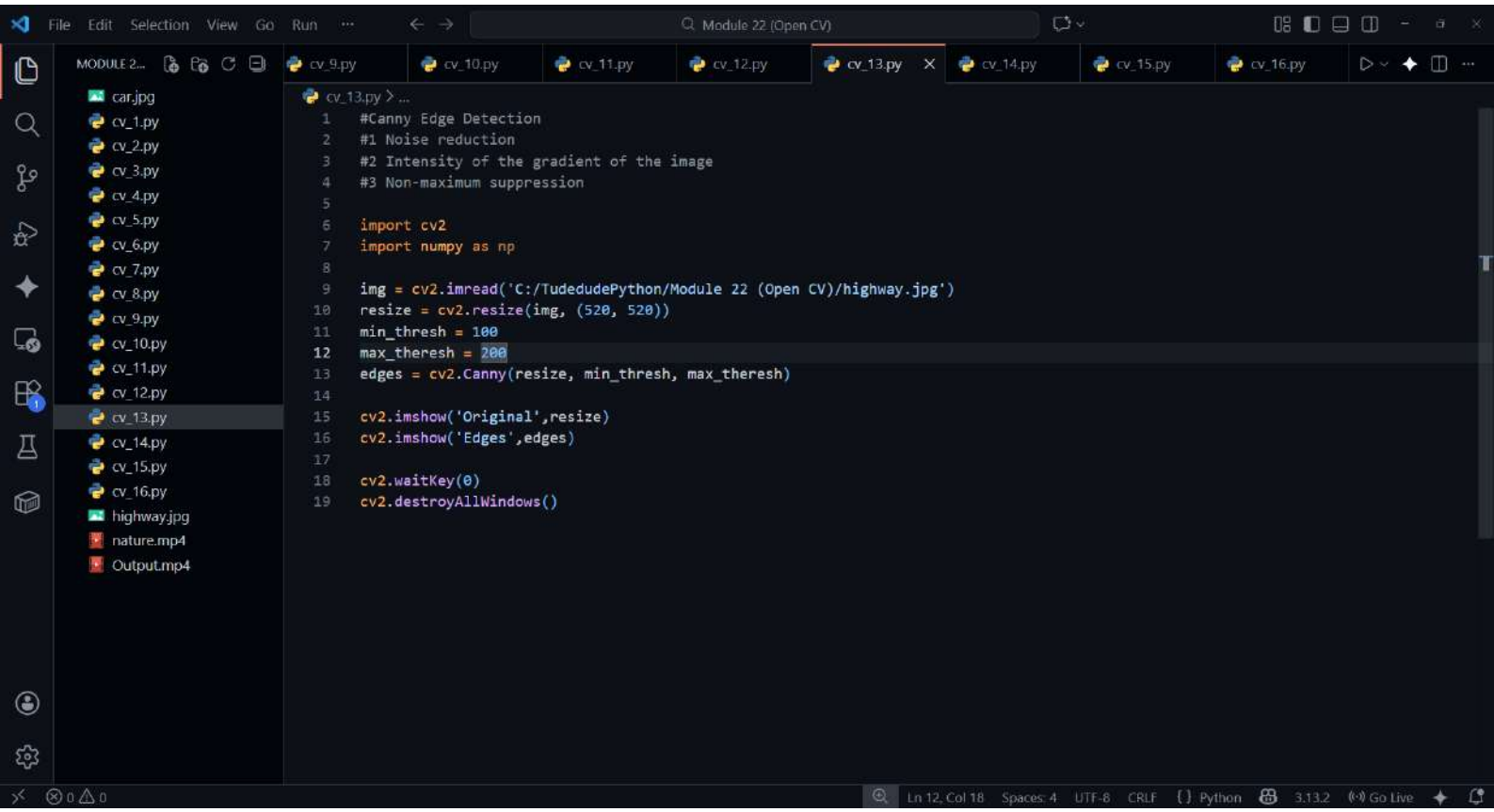
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_15.py > ...
import cv2
video =cv2.VideoCapture('C:/TudedudePython/Module 22 (Open CV)/nature.mp4')
fourcc = cv2.VideoWriter_fourcc(*'mp4v')
output = cv2.VideoWriter('Output.mp4', fourcc, 25.0, (4096, 2160))
while video.isOpened():
ret,frame = video.read()
if ret:
output.write(frame)
cv2.imshow('Frame',frame)
if cv2.waitKey(10) == ord('s'):
break
else:
break
cv2.destroyAllWindows()
Ln 13, Col 1 Spaces: 4 UTF-8 CRLF Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_14.py > ...
import cv2
video = cv2.VideoCapture('C:/TudedudePython/Module 22 (Open CV)/nature.mp4')
while video.isOpened():
ret, frame = video.read()
if not ret:
break
frame = cv2.resize(frame,(800,720))
cv2.imshow('Output',frame)
if cv2.waitKey(1) & 0xFF == ord('q'):
break
# cv2.waitKey(0)
cv2.destroyAllWindows()
Ln 13, Col 14 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
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cv_10.py
cv_11.py
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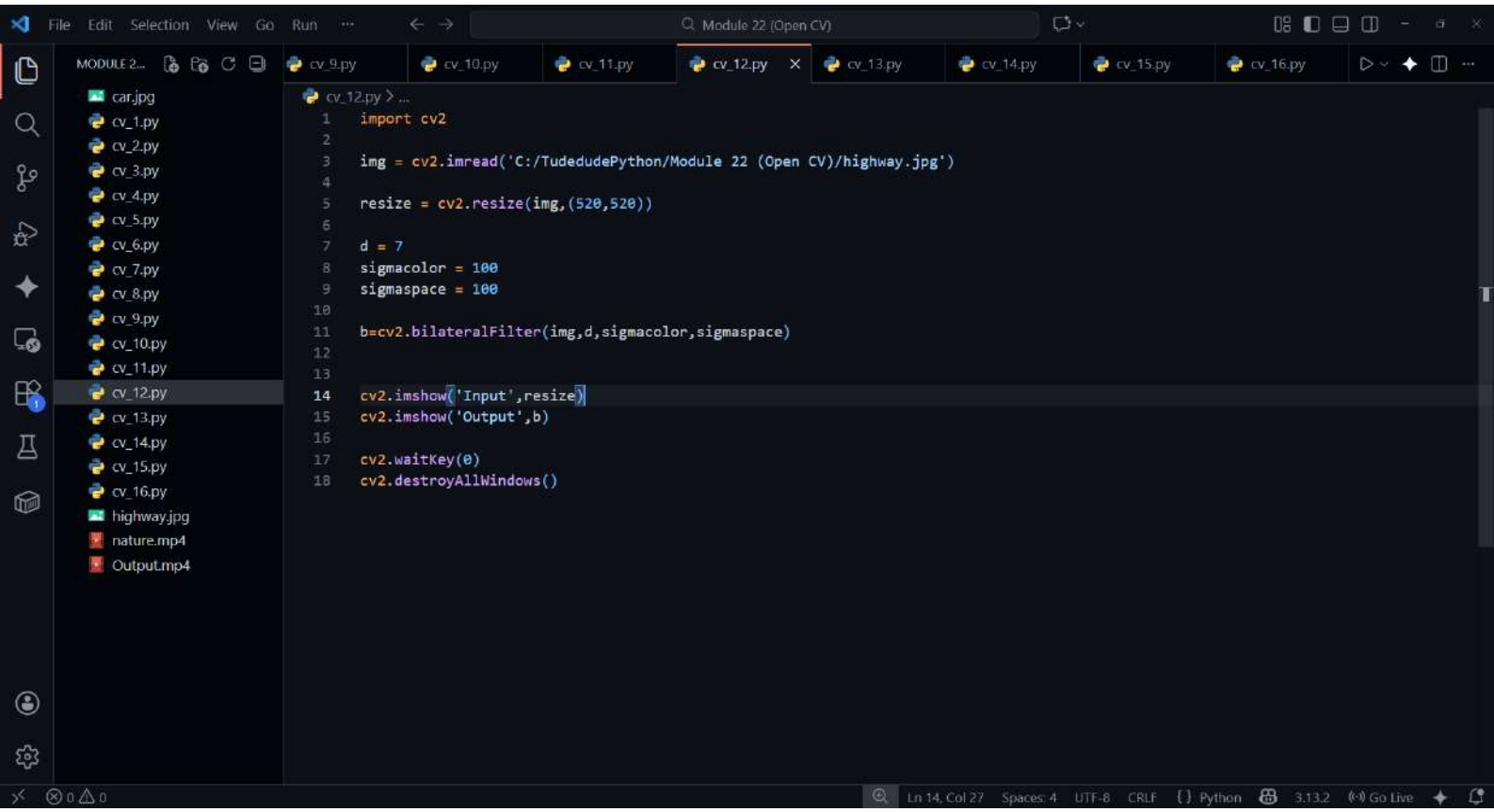
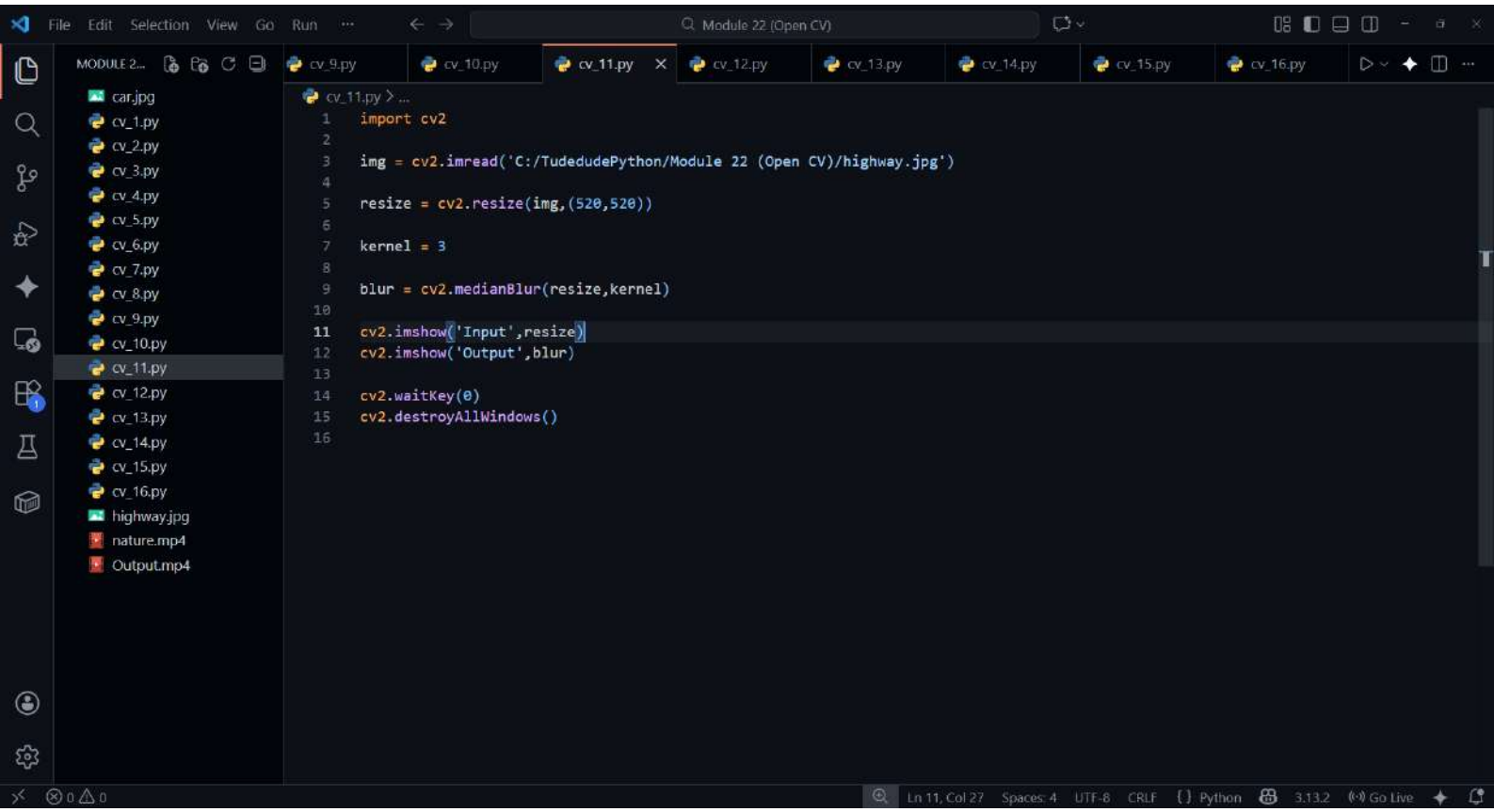
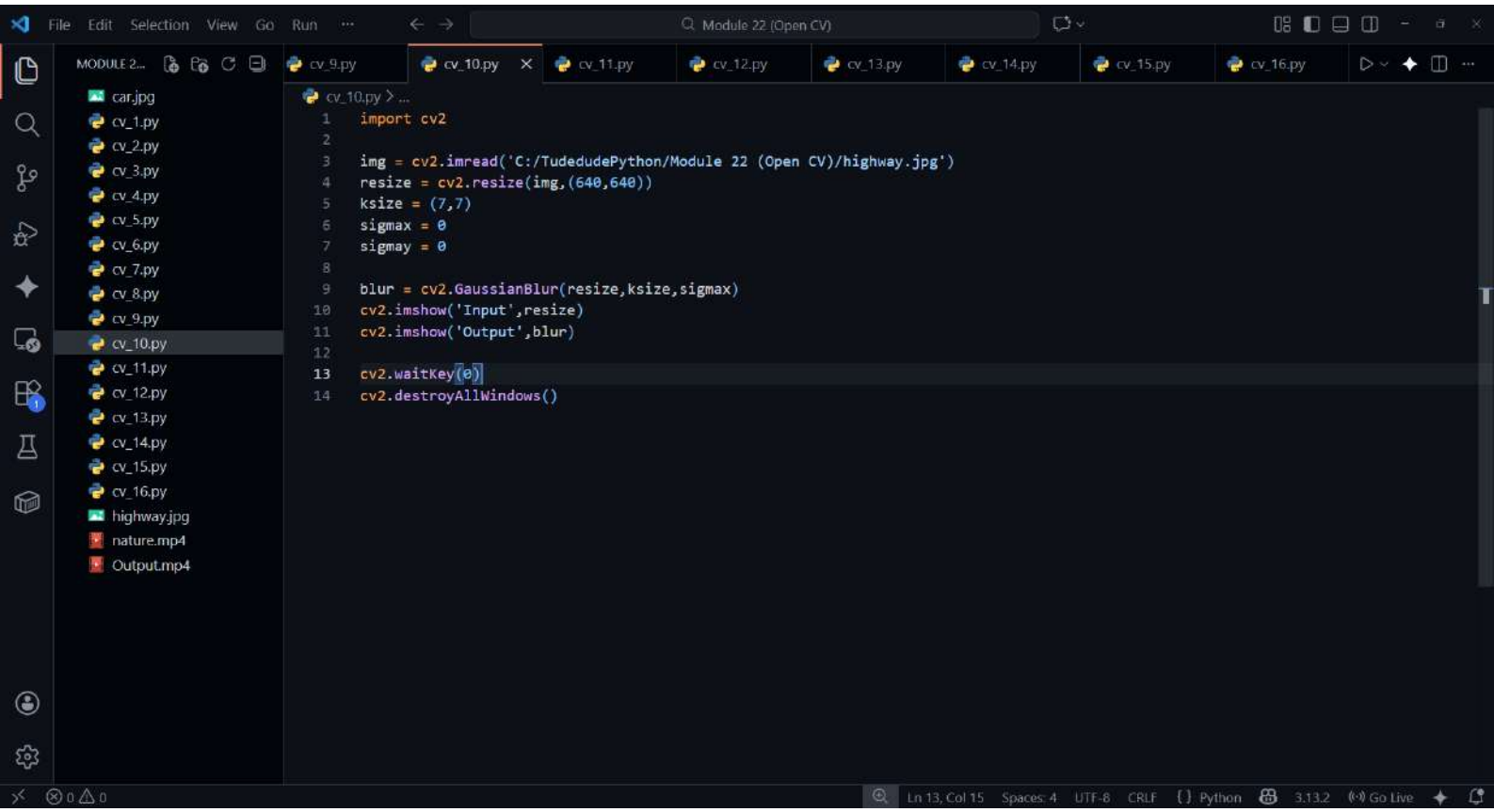
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_13.py > ...
#Canny Edge Detection
#1 Noise reduction
#2 Intensity of the gradient of the image
#3 Non-maximum suppression
import cv2
import numpy as np
img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
resize = cv2.resize(img, (520, 520))
min_thresh = 100
max_theresh = 200
edges = cv2.Canny(resize, min_thresh, max_theresh)
cv2.imshow('Original',resize)
cv2.imshow('Edges',edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln 12, Col 18 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_12.py > ...
1 import cv2
2
3 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
4
5 resize = cv2.resize(img,(520,520))
6
7 d = 7
8 sigmacolor = 100
9 sigmaspace = 100
10
11 b=cv2.bilateralFilter(img,d,sigmacolor,sigmaspace)
12
13
14 cv2.imshow('Input',resize)
15 cv2.imshow('Output',b)
16
17 cv2.waitKey(0)
18 cv2.destroyAllWindows()
Ln 14, Col 27 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_11.py > ...
1 import cv2
2
3 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
4
5 resize = cv2.resize(img,(520,520))
6
7 kernel = 3
8
9 blur = cv2.medianBlur(resize,kernel)
10
11 cv2.imshow('Input',resize)
12 cv2.imshow('Output',blur)
13
14 cv2.waitKey(0)
15 cv2.destroyAllWindows()
16
Ln 11, Col 27 Spaces: 4 UTF-8 CRLF Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
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cv_8.py
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cv_10.py
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cv_15.py
cv_16.py
highway.jpg
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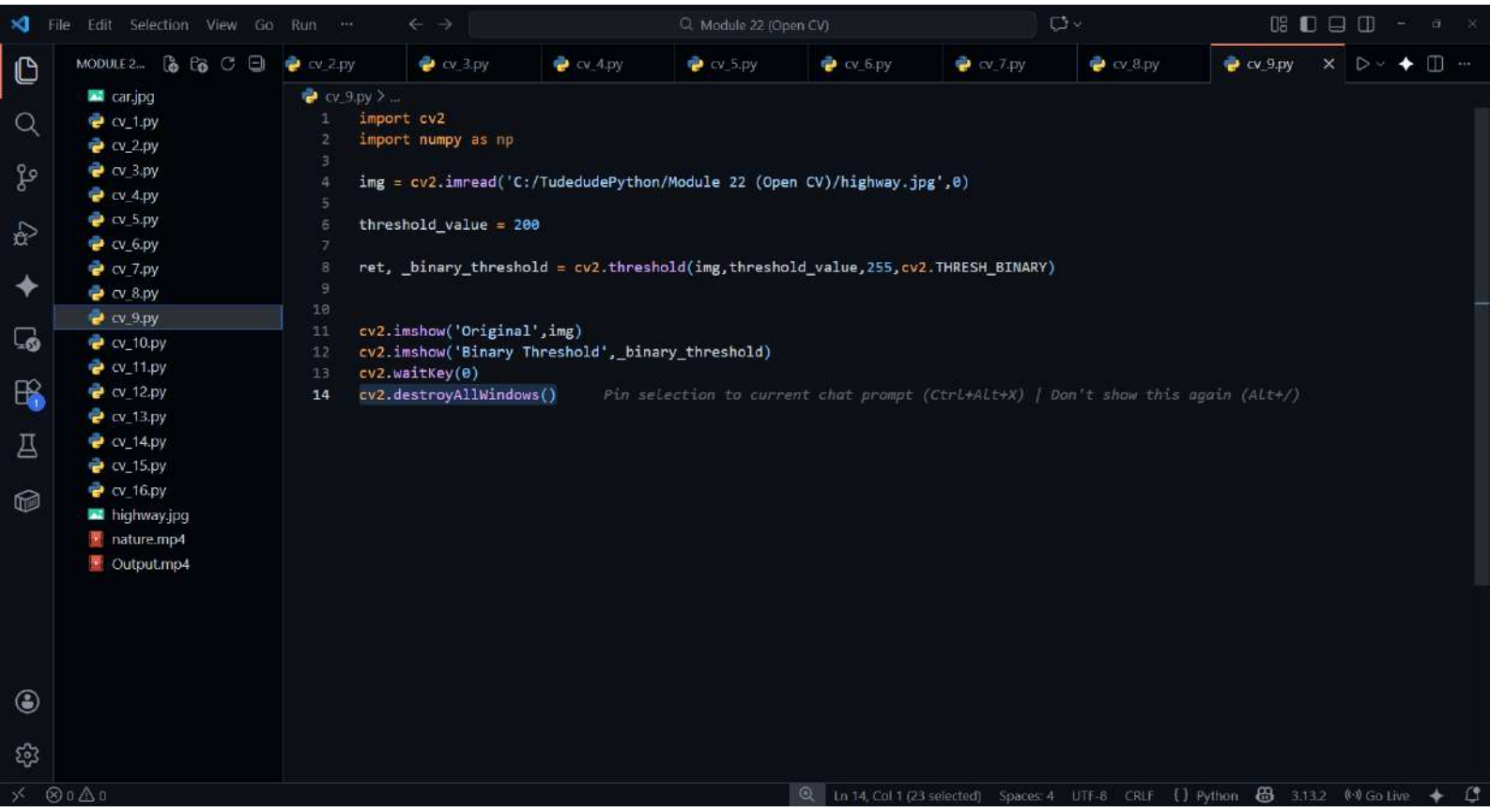
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
cv_10.py > ...
1 import cv2
2
3 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
4 resize = cv2.resize(img,(640,640))
5 ksize = (7,7)
6 sigmax = 0
7 sigmay = 0
8
9 blur = cv2.GaussianBlur(resize,ksize,sigmax)
10 cv2.imshow('Input',resize)
11 cv2.imshow('Output',blur)
12
13 cv2.waitKey(0)
14 cv2.destroyAllWindows()
Ln 13, Col 15 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
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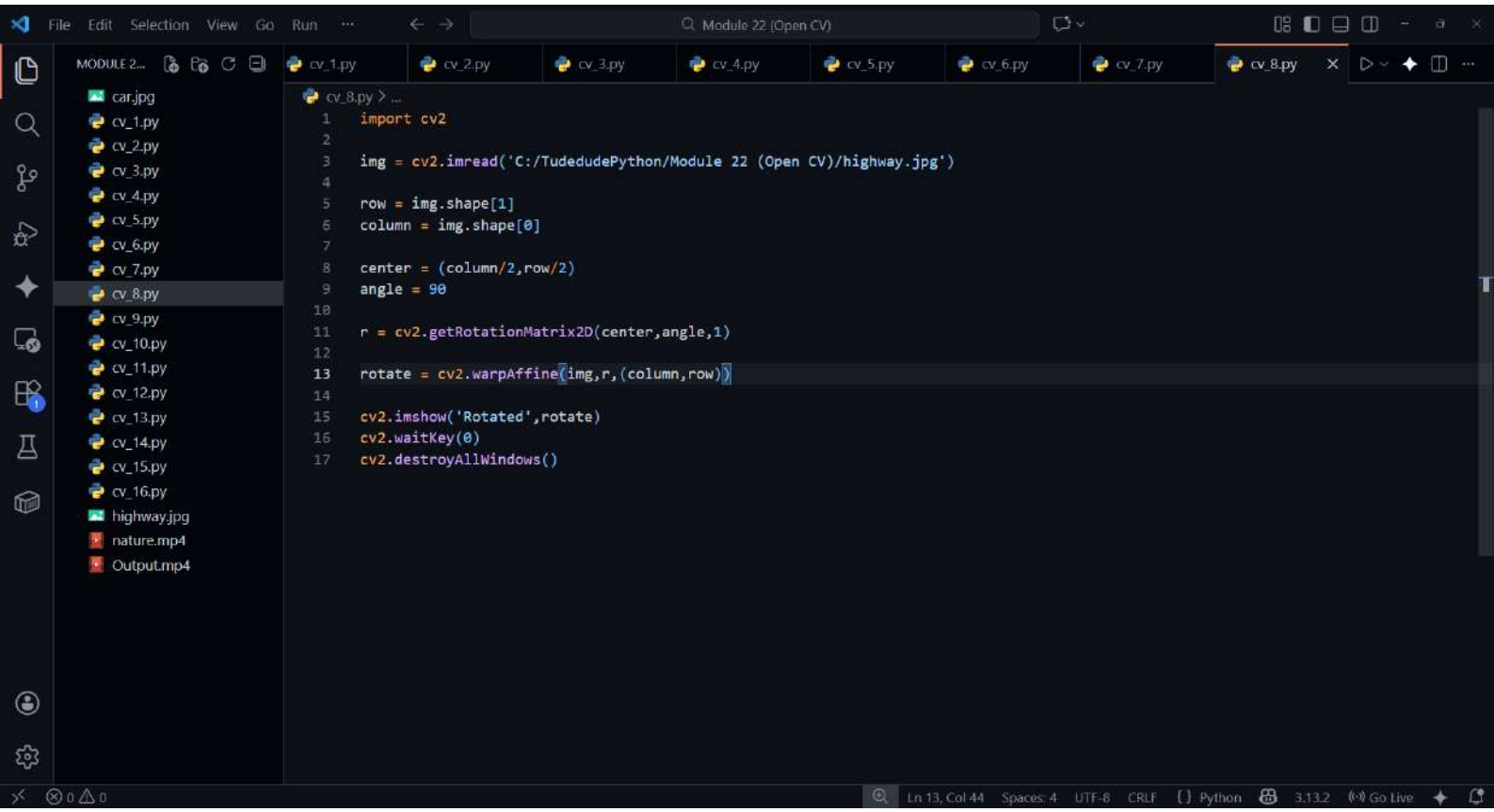
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_9.py > ...
1 import cv2
2 import numpy as np
3
4 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg',0)
5
6 threshold_value = 200
7
8 ret, _binary_threshold = cv2.threshold(img,threshold_value,255,cv2.THRESH_BINARY)
9
10
11 cv2.imshow('Original',img)
12 cv2.imshow('Binary Threshold',_binary_threshold)
13 cv2.waitKey(0)
14 cv2.destroyAllWindows()
Pin selection to current chat prompt (Ctrl+Alt+X) | Don't show this again (Alt+/)
Ln 14, Col 1 (23 selected) Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
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cv_11.py
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cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
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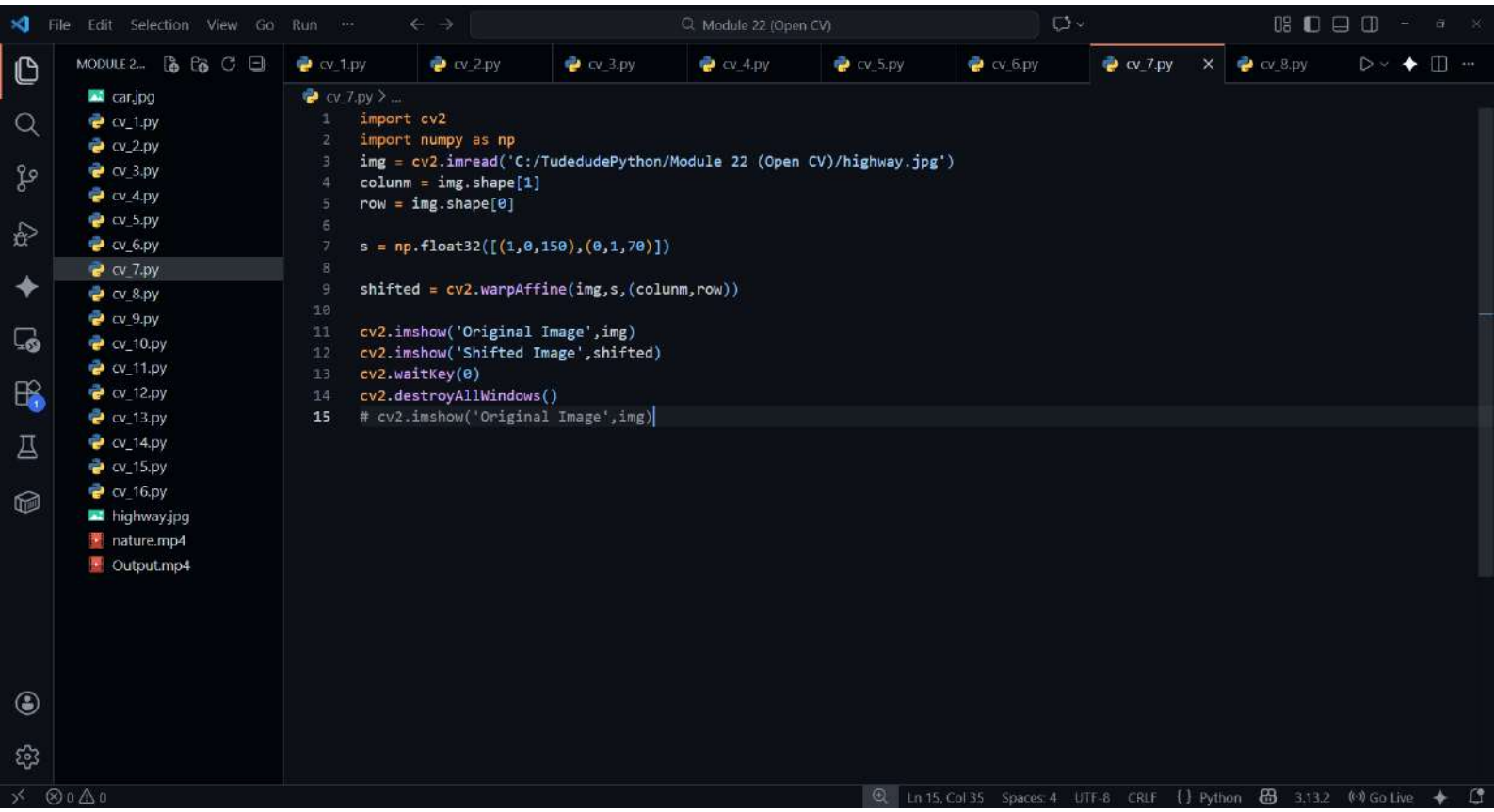
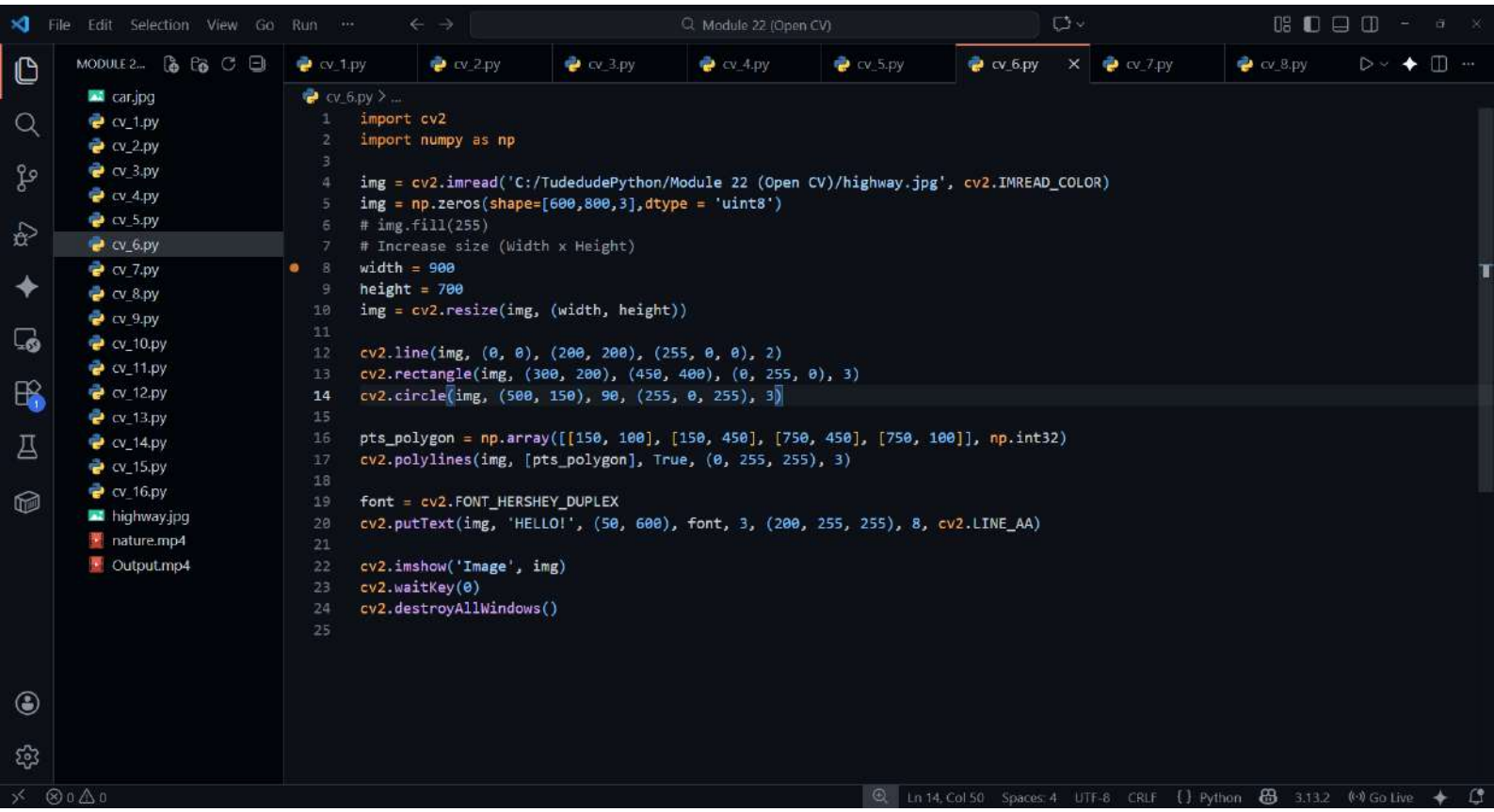
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_8.py > ...
1 import cv2
2
3 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
4
5 row = img.shape[1]
6 column = img.shape[0]
7
8 center = (column/2,row/2)
9 angle = 90
10
11 r = cv2.getRotationMatrix2D(center,angle,1)
12
13 rotate = cv2.warpAffine(img,r,(column,row))
14
15 cv2.imshow('Rotated',rotate)
16 cv2.waitKey(0)
17 cv2.destroyAllWindows()
Ln 13, Col 44 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_7.py > ...
1 import cv2
2 import numpy as np
3 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
4 columm = img.shape[1]
5 row = img.shape[0]
6
7 s = np.float32([(1,0,150),(0,1,70)])
8
9 shifted = cv2.warpAffine(img,s,(columm,row))
10
11 cv2.imshow('Original Image',img)
12 cv2.imshow('Shifted Image',shifted)
13 cv2.waitKey(0)
14 cv2.destroyAllWindows()
15 # cv2.imshow('Original Image',img)
0 0
Ln 15, Col 35 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
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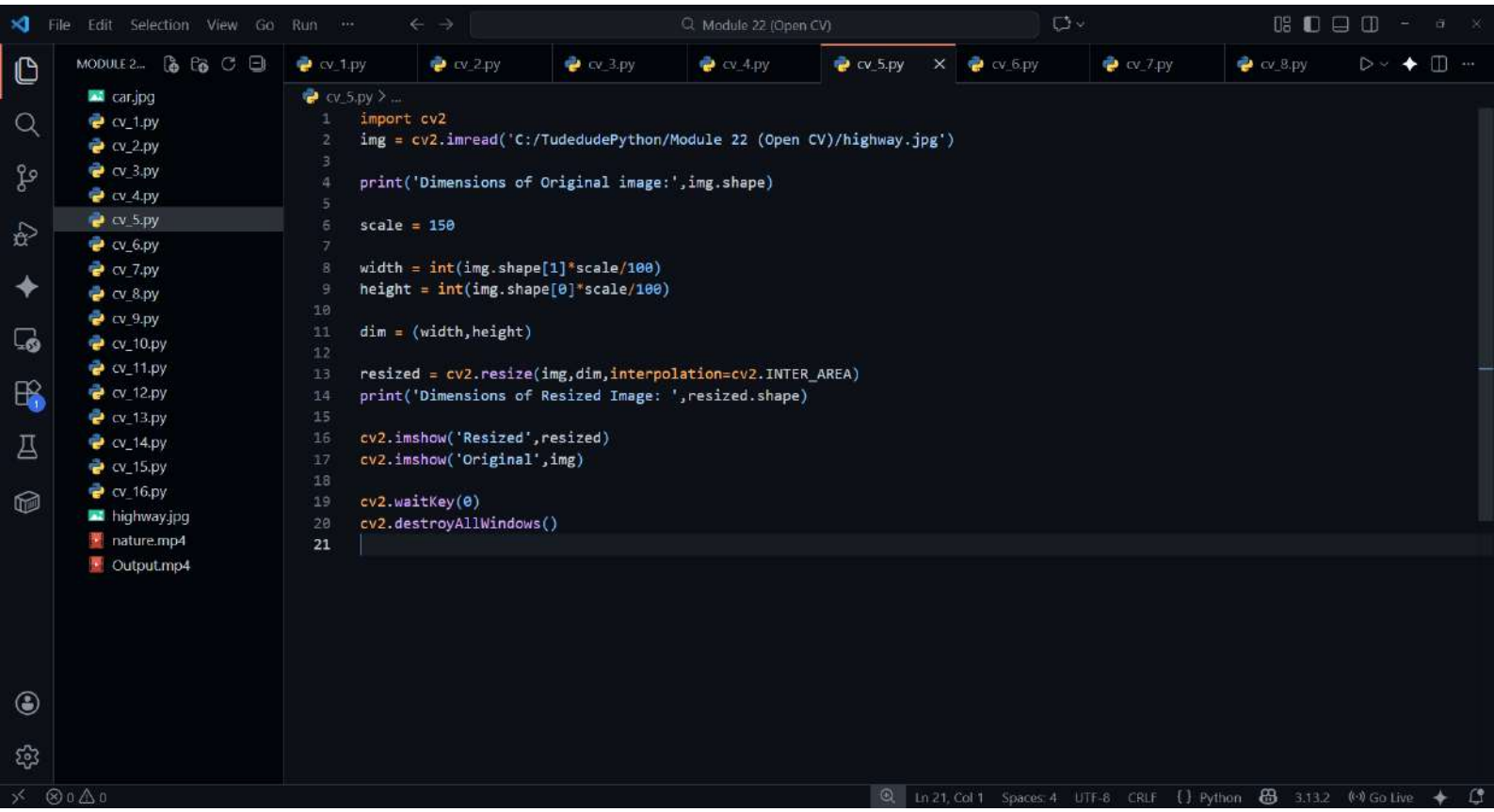
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_6.py > ...
1 import cv2
2 import numpy as np
3
4 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg', cv2.IMREAD_COLOR)
5 img = np.zeros(shape=[600,800,3],dtype = 'uint8')
6 # img.fill(255)
7 # Increase size (Width x Height)
8 width = 900
9 height = 700
10 img = cv2.resize(img, (width, height))
11
12 cv2.line(img, (0, 0), (200, 200), (255, 0, 0), 2)
13 cv2.rectangle(img, (300, 200), (450, 400), (0, 255, 0), 3)
14 cv2.circle(img, (500, 150), 90, (255, 0, 255), 3)
15
16 pts_polygon = np.array([[150, 100], [150, 450], [750, 450], [750, 100]], np.int32)
17 cv2.polylines(img, [pts_polygon], True, (0, 255, 255), 3)
18
19 font = cv2.FONT_HERSHEY_DUPLEX
20 cv2.putText(img, 'HELLO!', (50, 600), font, 3, (200, 255, 255), 8, cv2.LINE_AA)
21
22 cv2.imshow('Image', img)
23 cv2.waitKey(0)
24 cv2.destroyAllWindows()
25
Ln 14, Col 50 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
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cv_11.py
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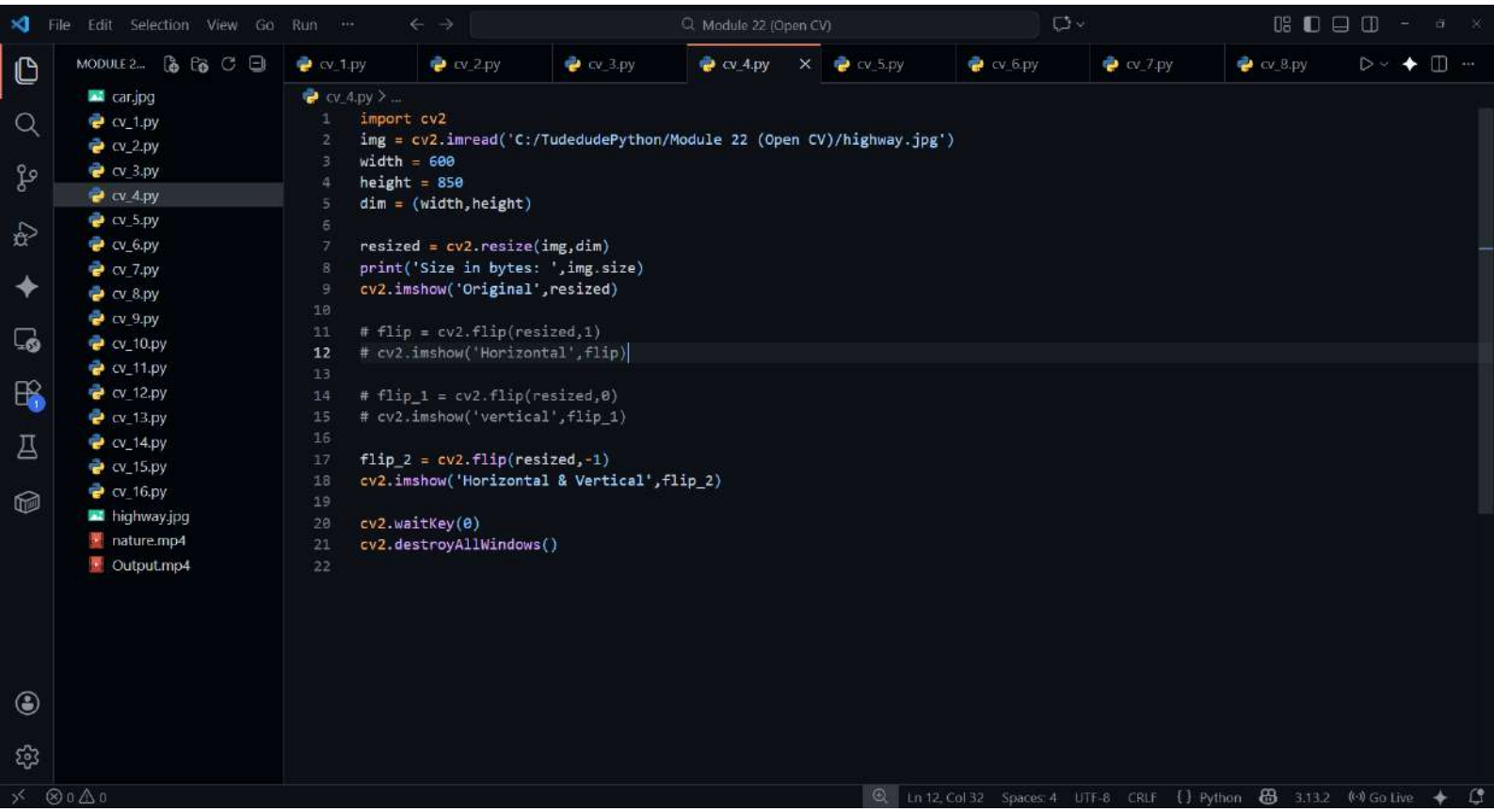
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_5.py > ...
1 import cv2
2 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
3
4 print('Dimensions of Original image:',img.shape)
5
6 scale = 150
7
8 width = int(img.shape[1]*scale/100)
9 height = int(img.shape[0]*scale/100)
10
11 dim = (width,height)
12
13 resized = cv2.resize(img,dim,interpolation=cv2.INTER_AREA)
14 print('Dimensions of Resized Image: ',resized.shape)
15
16 cv2.imshow('Resized',resized)
17 cv2.imshow('Original',img)
18
19 cv2.waitKey(0)
20 cv2.destroyAllWindows()
21
Ln 21, Col 1 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
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cv_4.py
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cv_15.py
cv_16.py
highway.jpg
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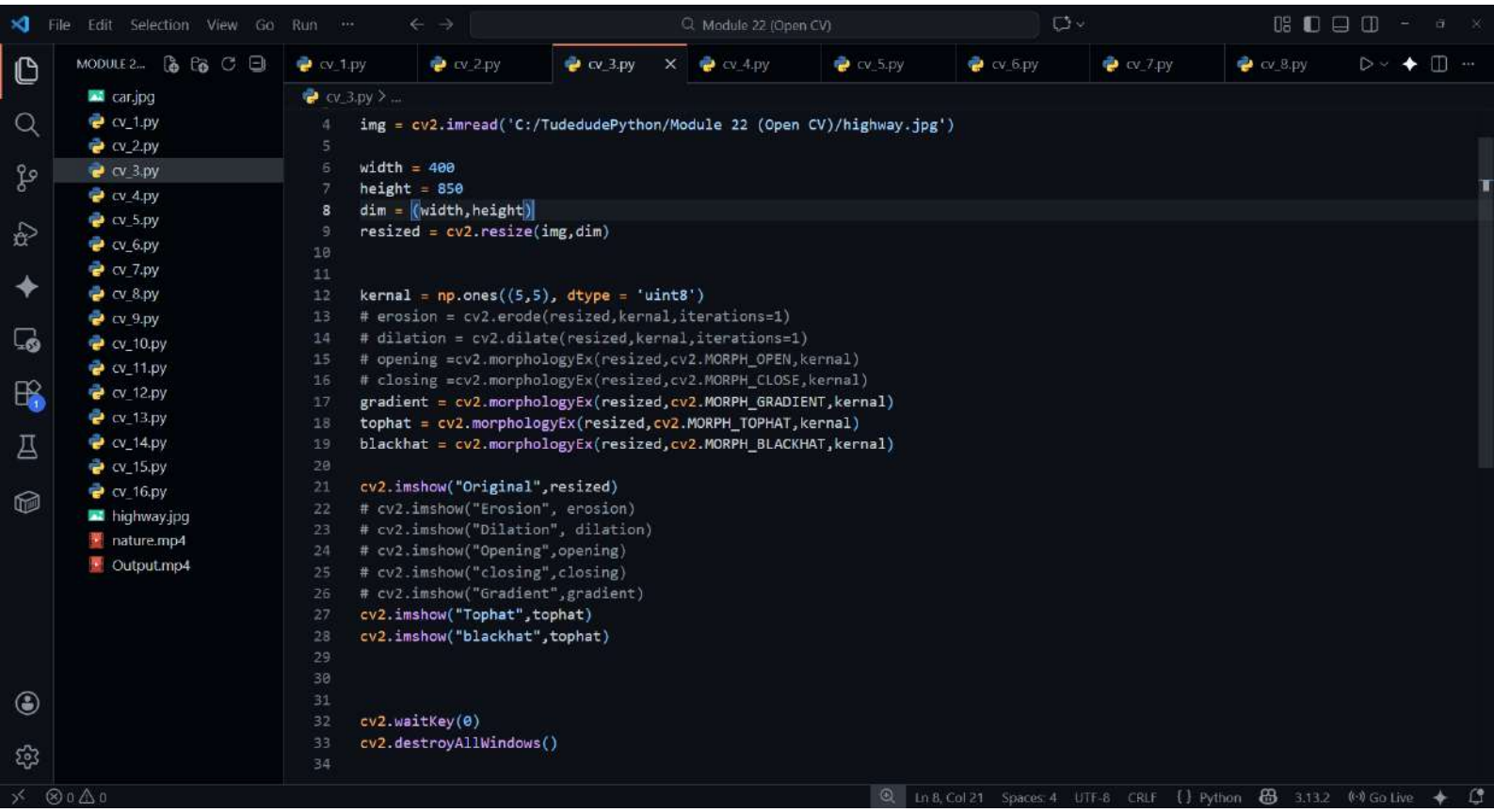
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_4.py > ...
1 import cv2
2 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
3 width = 600
4 height = 850
5 dim = (width,height)
6
7 resized = cv2.resize(img,dim)
8 print('Size in bytes: ',img.size)
9 cv2.imshow('Original',resized)
10
11 # flip = cv2.flip(resized,1)
12 # cv2.imshow('Horizontal',flip)
13
14 # flip_1 = cv2.flip(resized,0)
15 # cv2.imshow('vertical',flip_1)
16
17 flip_2 = cv2.flip(resized,-1)
18 cv2.imshow('Horizontal & Vertical',flip_2)
19
20 cv2.waitKey(0)
21 cv2.destroyAllWindows()
22
0 0
Ln 12, Col 32 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
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cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
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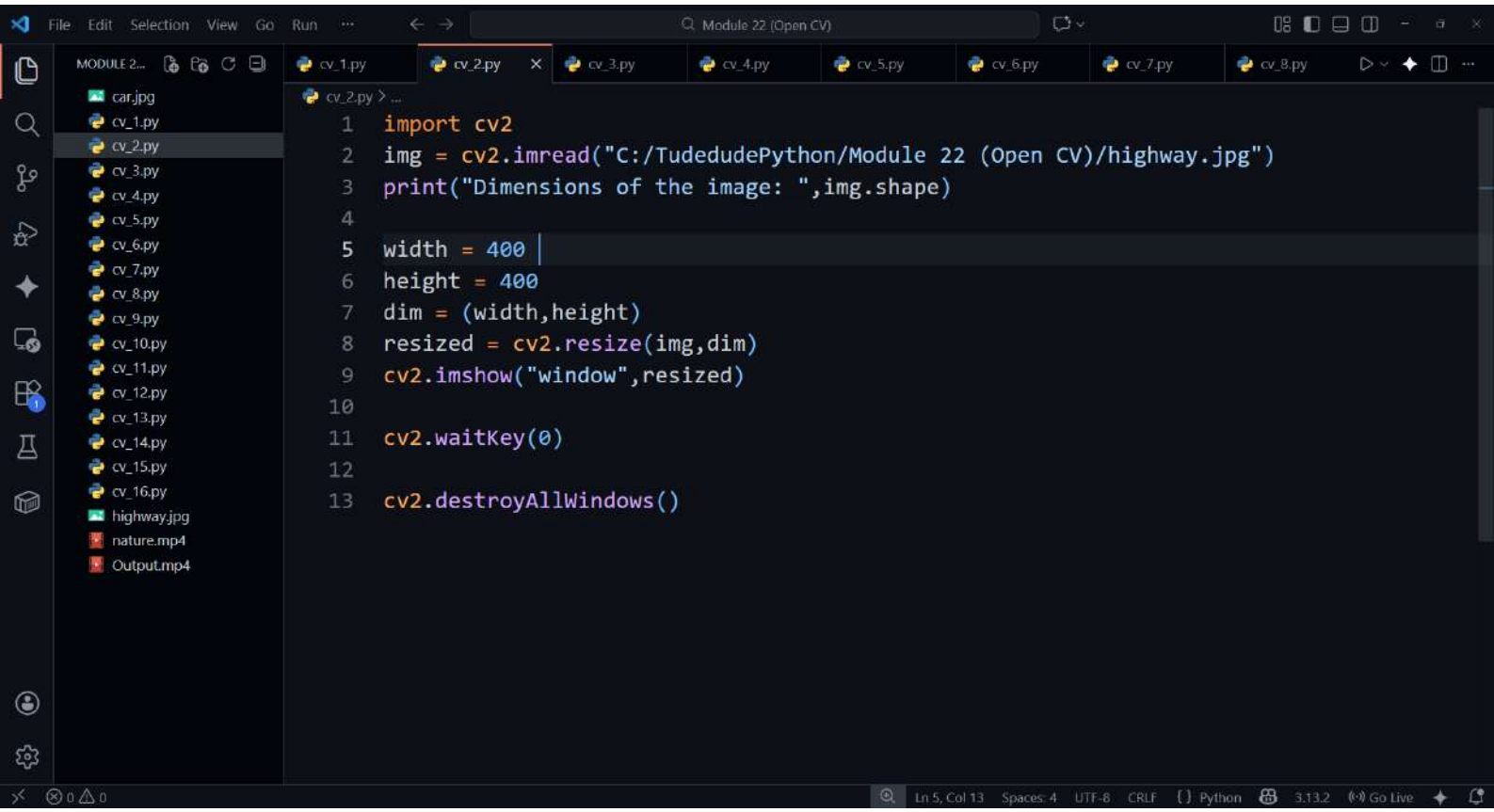
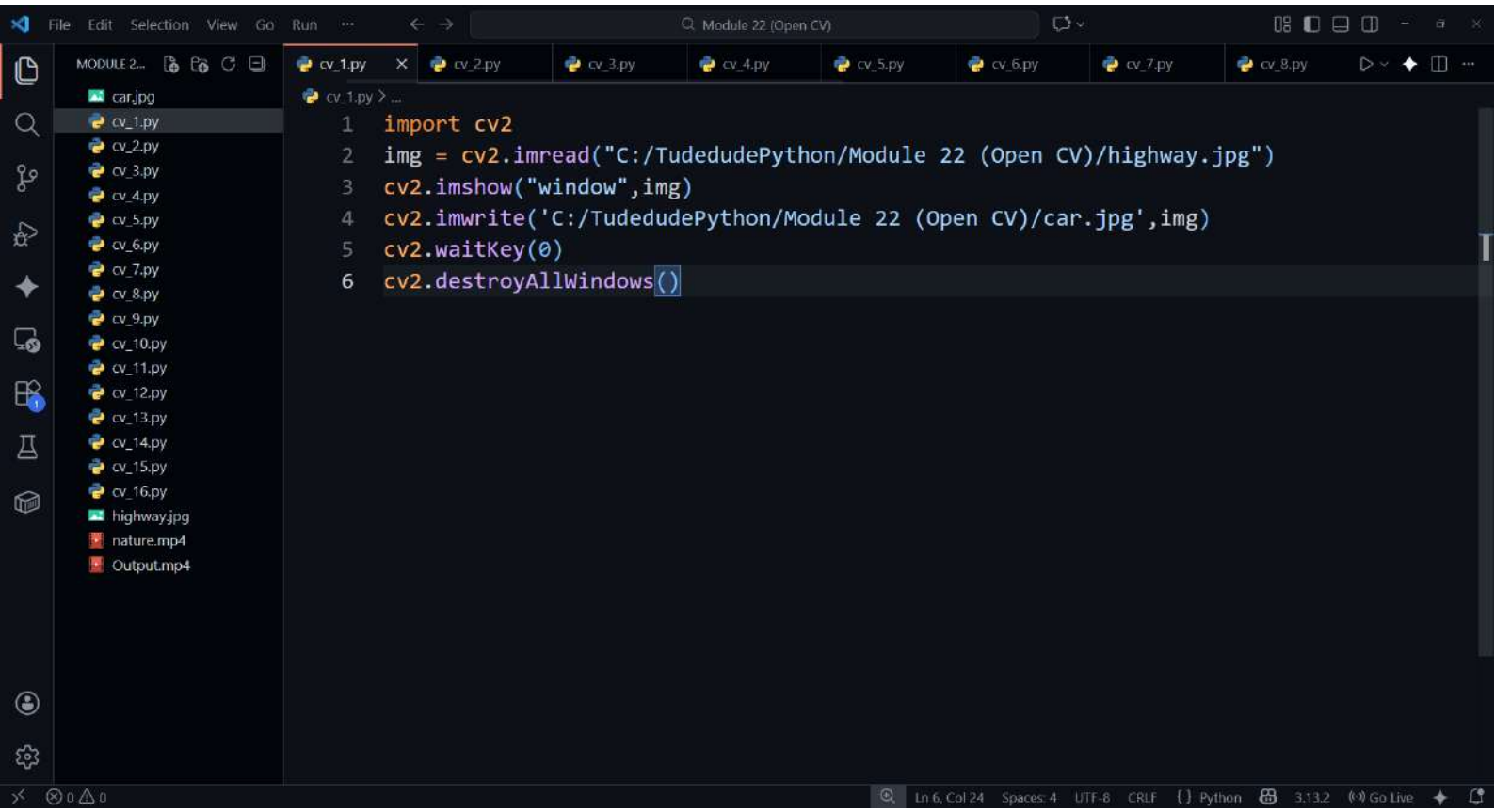
nature.mp4
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_3.py > ...
4 img = cv2.imread('C:/TudedudePython/Module 22 (Open CV)/highway.jpg')
5
6 width = 400
7 height = 850
8 dim = (width,height)
9 resized = cv2.resize(img,dim)
10
11
12 kernal = np.ones((5,5), dtype = 'uint8')
13 # erosion = cv2.erode(resized,kernal,iterations=1)
14 # dilation = cv2.dilate(resized,kernal,iterations=1)
15 # opening =cv2.morphologyEx(resized,cv2.MORPH_OPEN,kernal)
16 # closing =cv2.morphologyEx(resized,cv2.MORPH_CLOSE,kernal)
17 gradient = cv2.morphologyEx(resized,cv2.MORPH_GRADIENT,kernal)
18 tophat = cv2.morphologyEx(resized,cv2.MORPH_TOPHAT,kernal)
19 blackhat = cv2.morphologyEx(resized,cv2.MORPH_BLACKHAT,kernal)
20
21 cv2.imshow("Original",resized)
22 # cv2.imshow("Erosion", erosion)
23 # cv2.imshow("Dilation", dilation)
24 # cv2.imshow("Opening",opening)
25 # cv2.imshow("closing",closing)
26 # cv2.imshow("Gradient",gradient)
27 cv2.imshow("Tophat",tophat)
28 cv2.imshow("blackhat",tophat)
29
30
31
32 cv2.waitKey(0)
33 cv2.destroyAllWindows()
34
Ln 8, Col 21 Spaces: 4 UTF-8 CRLF Python 3.13.2 Go Live


File Edit Selection View Go Run ···
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_9.py
cv_10.py
cv_11.py
cv_12.py
cv_13.py
cv_14.py
cv_15.py
cv_16.py
highway.jpg
nature.mp4
Output.mp4
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_2.py > ...
1 import cv2
2 img = cv2.imread("C:/TudedudePython/Module 22 (Open CV)/highway.jpg")
3 print("Dimensions of the image: ",img.shape)
4
5 width = 400
6 height = 400
7 dim = (width,height)
8 resized = cv2.resize(img,dim)
9 cv2.imshow("window",resized)
10
11 cv2.waitKey(0)
12
13 cv2.destroyAllWindows()
Ln 5, Col 13 Spaces: 4 UTF-8 CRLF { } Python 3.13.2 Go Live


File Edit Selection View Go Run
Module 22 (Open CV)
MODULE 2...
car.jpg
cv_1.py
cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
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cv_15.py
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cv_2.py
cv_3.py
cv_4.py
cv_5.py
cv_6.py
cv_7.py
cv_8.py
cv_1.py > ...
1 import cv2
2 img = cv2.imread("C:/TudedudePython/Module 22 (Open CV)/highway.jpg")
3 cv2.imshow("window",img)
4 cv2.imwrite('C:/TudedudePython/Module 22 (Open CV)/car.jpg',img)
5 cv2.waitKey(0)
6 cv2.destroyAllWindows()
Ln 6, Col 24
Spaces: 4
UTF-8
CRLF
Python
3.13.2
Go Live